

Cybersecurity Awareness and Digital Safety Practices Among Adolescents Aged 16–20 in Dhangadhi, Nepal



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ABSTRACT

The rapid growth of internet access and smartphone usage has transformed how adolescents communicate, learn, and interact. Social media, messaging applications, and online platforms are now integral to daily life, offering educational and social opportunities but also exposing users to cybersecurity risks such as phishing, malware, scams, identity theft, and privacy breaches.

This study assessed cybersecurity awareness and digital safety practices among adolescents aged 16–20 in Dhangadhi, Nepal. A descriptive research design was employed, with data initially collected from more than 395 participants through structured questionnaires distributed across six colleges in Dhangadhi via both physical visits and online platforms. After data cleaning to remove incomplete or invalid responses, a final sample of 372 valid responses was retained for analysis. The sample comprised 193 males (51.9%) and 178 females (47.8%), with the majority aged 17 years (42.7%). Regarding internet usage, 43.5% of respondents used the internet 2–4 hours daily, while 12.6% reported using it for more than 6 hours per day.

Preliminary findings indicate that while awareness of certain threats is relatively high 91.6% recognized the importance of strong passwords and 75.7% were aware of public Wi-Fi risks significant gaps remain. Only 44.3% of respondents felt confident spotting suspicious messages, and only 53.5% reported knowledge and use of Two-Factor Authentication (2FA). In terms of digital safety practices, 86.3% claimed to use strong passwords and 78.7% reported not sharing OTP or sensitive information; however, only 37.6% used antivirus software or a VPN, and 50.7% regularly updated their software and applications. Concerning risky behaviors, 47.8% admitted to ignoring cybersecurity risks even when aware of them, highlighting a critical gap between knowledge and practice. Notably, 85.1% of respondents agreed that cybersecurity awareness is important, and 49.5% reported that cyber incidents had affected them or someone they know.

Keywords: Cybersecurity Awareness, Digital Literacy, Online Safety, Cybercrime, Adolescents, Nepal

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Chapter 1

INTRODUCTION

1.1 Background of the study

The internet and smartphones have become an important part of adolescents' daily lives. Young people use digital technologies for communication, education, entertainment, and social networking. While these technologies provide many opportunities, they also expose users to various cybersecurity threats such as phishing, online scams, malware, identity theft, and privacy breaches. Due to their high level of online activity, adolescents are often more vulnerable to these risks.

In Nepal, the use of the internet and social media among young people has increased rapidly in recent years. However, awareness of cybersecurity and safe online practices has not grown at the same pace. Many students may have basic knowledge of digital technologies but lack the skills needed to recognize cyber threats and protect themselves online.

Dhangadhi is no exception to this trend. Students increasingly depend on digital platforms for learning, communication, and entertainment, yet little is known about their level of cybersecurity awareness and actual online safety practices. Understanding these aspects is important for developing effective educational programs and promoting responsible digital behavior among adolescents.

This study aims to assess cybersecurity awareness and digital safety practices among adolescents aged 16–20 years in Dhangadhi, Nepal. Data were collected from 372 Grade 11 and 12 students studying in both public and private institutions. The participants represented different academic streams, including Computer Science, Bio Science, Nursing, Pharmacy, and Management. The survey was conducted in six educational institutions: Aishwarya Vidya Niketan (AVN), Alliance International Model School, Kailali Model School, Panchodaya Higher Secondary School, Navajeevan College of Nursing and Health Science, and NAST Secondary School.

The findings of this study are expected to provide insights into adolescents' understanding of cybersecurity issues and their online safety behaviors. The results may help educators, policymakers, and schools identify existing gaps and develop strategies

to improve cybersecurity awareness and encourage safer digital practices among young people.

1.2 Literature Review

(Bhattacharya et al., 2025) refers that, Cybersecurity safeguards systems, networks, and programs to prevent digital attacks that seek to gain unauthorized access, manipulate, or compromise sensitive data. (Wongmahesak et al., 2025) enhances digital literacy and reduce cybercrime fear, emphasizing customized educational approaches for various age groups, including community workshops, school integration, and online resources.

(Nayyar & Gupta) highlights the growing importance of cybersecurity in education as digital tools and virtual learning environments become more widespread. It emphasizes the need for cyber awareness, secure practices, and integrating cybersecurity education into the curriculum to protect students and institutions from emerging threats. Overall, it stresses building a culture of safety to ensure a secure digital future for learning. (Hussain & Phulpoto, 2024) examines how digital literacy empowers individuals across different groups through surveys, interviews, and literature analysis. Findings show that higher digital literacy improves job opportunities, social participation, and overall digital engagement while highlighting challenges like access, skills gaps, and the digital divide. The research also proposes training programs and policy support to strengthen digital literacy in communities.

(Prakash & Sanju, 2024) evaluates how effectively social media can be used to promote cyber safety awareness by analyzing public knowledge, attitudes, and online behaviors. It also examines demographic factors to determine how social media campaigns can be better designed for different groups.

(Bayzan et al.) examines youth-focused digital literacy programs and assessment tools in Canada to address growing concerns about online safety, privacy, and cyber risks. An environmental scan identified 15 existing programs, while a literature review found 12 assessment tools. Based on these findings, a new adaptable digital literacy program and a 15-item assessment tool were developed to strengthen youth skills in digital fluency, safety, ethics, and consumer awareness. The research highlights the need for targeted digital literacy education to better prepare youth for the digital world.

Cybercrime has emerged as a major threat in the digital era, affecting individuals, organizations, and governments worldwide. As digital dependence increases, users with limited cybersecurity awareness and digital literacy become more vulnerable to cyber threats. Understanding the nature of cybercrime is therefore essential for promoting safe digital practices. (Kaspersky, 2025) defines cybercrime as criminal activities that target or use computers, networks, or digital devices, primarily for financial gain, data theft, or system disruption. The study categorizes cybercrime into various forms, including phishing, identity theft, malware attacks, ransomware, cyberextortion, and distributed denial-of-service (DDoS) attacks. It highlights that cybercriminals often exploit users' lack of awareness by using deceptive techniques such as fraudulent emails, fake websites, and social engineering tactics. The report further emphasizes that cybercrime is increasingly affecting not only large organizations but also individuals and small businesses, leading to financial loss, reputational damage, and loss of trust in digital systems. Additionally, it stresses that the rise in cybercrime is closely linked to insufficient cybersecurity awareness and poor digital practices among users. The study recommends adopting preventive measures such as using strong passwords, avoiding suspicious links, keeping software updated, and protecting personal information online. These recommendations align with the objectives of the present study, as they highlight the role of cybersecurity awareness and digital literacy in reducing cyber risks and encouraging safe digital behavior. The findings support the need for educational and awareness-based interventions to empower users to protect themselves from cybercrime in an increasingly digital society.

Children represent one of the most vulnerable groups in the digital world. According to (Innocenti, 2021a), children increasingly use the internet for playing, learning, and socializing. However, this exposure brings risks such as cyberbullying, online predators, and misinformation. UNICEF emphasizes that digital literacy from an early age is essential to balance opportunities with risks. Similarly, the UNICEF (Innocenti, 2021b) found that while children benefit from online learning and communication, weak cybersecurity measures expose them to privacy breaches, exploitation, and harassment.

Teenagers face distinct challenges as they navigate social media and online communities. Information leaks often result in harassment and bullying, with long-term psychological impacts. A study by (Sun et al., 2025) examined how cyber moral literacy

influences cyberbullying behaviors among late adolescents. The research explored three critical questions:

- a) How does cyber moral literacy influence cyberbullying behaviors?
- b) To what extent does moral disengagement mediate the relationship between cyber moral literacy and cyberbullying?
- c) How does guilt moderate the relationship between cyber moral literacy and cyberbullying?

Findings suggest that teenagers with stronger moral literacy are less likely to engage in cyberbullying, while moral disengagement increases harmful behaviors. Guilt acts as a moderating factor, reducing the likelihood of harassment when adolescents recognize the ethical implications of their actions. This underscores the importance of integrating moral and ethical dimensions into cybersecurity education for teenagers.

Adults, though generally more aware of the consequences of unsafe practices, often ignore them due to convenience or affordability issues. The use of pirated products, whether entertainment or productivity software, poses serious risks. It is found that pirated software often contains hidden malware, spyware, or backdoors that compromise personal and organizational data, exposing users to ransomware attacks and system instability. Similarly, the (Partida, 2020) reported that cracked or counterfeit software bypasses security protocols, making systems more vulnerable to cyberattacks. Despite awareness, many adults continue to rely on pirated products, highlighting the gap between knowledge and practice.

The study “Perceived Digital Threats Influencing Smartphone Use Among the Aging Population” by R. Kanimozhi, V. Padmavathi, and P. S. Ramesh highlights a critical issue: many older adults are not sufficiently prepared to use modern technology with confidence. As the world rapidly digitizes, the global rise in the elderly population has been accompanied by increasing adoption of smartphones. These devices are not just tools of convenience, they help seniors maintain relationships, access healthcare, and engage in recreational activities, ultimately improving their overall quality of life. However, the research reveals a concerning trend. A significant number of elderly individuals are stepping into the digital world without fully understanding the risks and challenges of cybersecurity. While smartphones open doors to connection and

independence, they also expose older adults to threats such as privacy breaches, fraud, and data misuse.(Kanimozhi et al., 2025)

Cybersecurity awareness plays a crucial role in reducing cybercrime and promoting safe digital practices, particularly among general users. In developing countries like Nepal, the lack of awareness regarding online threats significantly increases users' vulnerability to cyber fraud, identity theft, and online scams. The (NepalPolice, 2021) highlights the importance of adopting basic cyber safety practices to prevent cybercrime. The guidelines emphasize avoiding suspicious links and emails, not sharing personal information or private photos online, and being cautious of fraudulent job offers and online scams. The Bureau also stresses the use of strong passwords, regular password changes, and enabling multi-factor authentication (MFA) to protect online accounts. Additionally, users are advised to avoid conducting sensitive financial transactions on public Wi-Fi networks and to regularly update devices with reliable antivirus software. Furthermore, the Cyber Bureau underscores the importance of data protection through regular backups and encourages families to educate children and household members about online risks. These preventive strategies demonstrate that cybercrime prevention is not solely dependent on technical security systems but also on user awareness and responsible digital behavior. This source is particularly relevant to the present study as it provides a local and practical perspective on cybersecurity awareness, supporting the need for educational initiatives to promote safe online practices and reduce cybercrime in Nepal.

1.3 Objectives

1.3.1 General Objective

- To assess cybersecurity awareness and digital safety practices among adolescents aged 16–20 in Dhangadhi, Nepal.

1.3.2 Specific Objectives

- To measure the level of cybersecurity awareness among adolescents aged 16–20.
- To identify common cybersecurity threats encountered by adolescents on digital platforms.

- To examine the relationship between digital literacy and cybersecurity awareness among adolescents.
- To analyze the digital safety practices adopted by adolescents to prevent or respond to cybersecurity threats.
- To provide recommendations for improving cybersecurity education and promoting safe digital behavior among adolescents.

1.4 Hypothesis

H1: Adolescents with higher digital literacy demonstrate greater awareness of cybersecurity threats compared to adolescents with lower digital literacy.

H2: Adolescents who have greater awareness of cybersecurity threats are more likely to adopt safe digital practices.

H3: Adolescents who spend more time online or use social media frequently are more likely to be exposed to cybersecurity risks.

1.5 Conceptual Framework

The conceptual framework of this study illustrates the relationships between adolescents' demographic factors, digital exposure, cybersecurity awareness, and digital safety practices. It assumes that several independent variables such as age, gender, education level, time spent online, social media usage, digital literacy, and previous experience with online threats; can influence adolescents' understanding of cybersecurity.

Cybersecurity awareness acts as a mediating variable connecting these independent factors to the outcome variable, digital safety practices. Adolescents with higher cybersecurity awareness are more likely to adopt safe online behaviors, including protecting personal information, avoiding suspicious links, reporting scams, and using secure platforms.

By analyzing these relationships, the study aims to identify which factors most strongly predict cybersecurity awareness and digital safety practices among adolescents aged 16–20 in Dhangadhi, Nepal. This framework provides the basis for understanding how personal and digital exposure factors interact with awareness to shape safe online behavior.

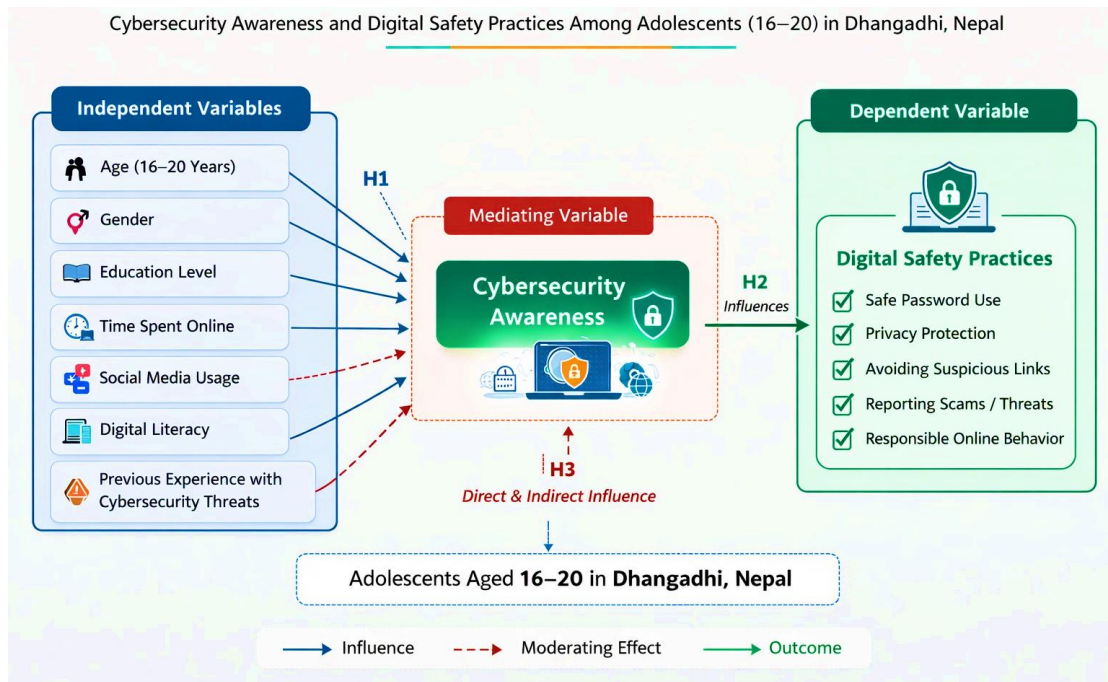


Figure 1.1 Conceptual Framework

Chapter 2

METHODOLOGY

2.1 Description of the Research Station

This study was conducted with the support of the National Academy of Science and Technology (NAST), Dhangadhi, which provided essential resources throughout the research process. The institution offered access to computers, reliable internet connectivity, and a dedicated workspace for preparing questionnaires, managing responses, and analyzing data. Continuous guidance from supervisors ensured that the study maintained academic standards and methodological rigor at every stage of the research.

2.2 Research Design, Sample Size, and Sampling Strategies

A descriptive research design was employed to investigate cybersecurity awareness and digital safety practices among adolescents aged 16–20 in Dhangadhi, Nepal.

2.2.1 Research Design

A descriptive research design was chosen as it is most appropriate for assessing the current state of cybersecurity awareness and digital safety practices within a defined population. This design allowed the researchers to systematically collect, organize, and analyze data without manipulation of variables, ensuring an accurate and objective representation of adolescents' knowledge, attitudes, and online behaviors.

2.2.2 Sample Size

The target sample size was 384 participants, calculated using Cochran's formula for an infinite population at a 95% confidence level and a 5% margin of error. A total of 395 responses were collected across six colleges in Dhangadhi through direct physical visits to each institution. Following data cleaning, 372 responses were found to be valid and complete, achieving 96.9% of the target sample size. The remaining 23 responses were excluded due to incomplete entries, duplicate submissions, or respondents falling outside the 16–20 age range.

2.2.3 Data Collection Methods

Structured questionnaires were administered through the following channels:

- Physical visits to six colleges in Dhangadhi

- Direct in-person interactions with students during college visits
- Community interactions within college premises

2.2.4 Sampling Strategies

Convenience Sampling: Participants who were readily accessible during college visits and community interactions were engaged directly, allowing efficient data collection within the study timeline.

Purposive Sampling: Only adolescents within the 16–20 age range were selected to ensure the relevance and focus of responses. Respondents outside this age range were excluded during the data cleaning stage.

This combined sampling approach ensured the inclusion of adolescents from diverse educational backgrounds, faculties, and levels of online experience, resulting in a comprehensive and representative dataset.

2.3 Study Area

The study was conducted in Dhangadhi, Kailali, a rapidly developing urban center in western Nepal where adolescent internet usage has increased significantly in recent years. Data was collected through physical visits to the following six colleges:

Table 2.1
Data Collections Modules

S.No .	College	Respondents Collected	Valid Respondents	Percentage
1.	NAST Secondary School	80	76	20.4%
2.	Aishwarya Vidya Niketan	75	71	19.1%
3.	Alliance International Model School	73	70	18.8%
4.	Navajeevan College of Nursing and Health Science	60	57	15.3%
5.	Panchodaya Higher Secondary School	60	57	15.3%
6.	Kailali Model College	47	41	11.0%
Total		395	372	100%

2.4 Research Tools, Techniques, and Data Collection Strategies

Data was collected using a structured questionnaire specifically designed to assess cybersecurity awareness and digital safety practices among adolescents aged 16–20. The questionnaire consisted of 41 items organized into the following sections:

- **Section A:** Demographic information (age, gender, college, faculty, scheme)
- **Section B:** Internet usage patterns (daily usage, social media frequency, preferred platforms)
- **Section C:** Cybersecurity awareness (knowledge of threats, phishing, 2FA, public Wi-Fi risks, suspicious messages)
- **Section D:** Digital safety practices (password behavior, OTP sharing, link avoidance, software updates, antivirus use)
- **Section E:** Risky behaviors and cyber threat exposure (past incidents, feeling unsafe online, ignoring known risks)

Awareness and practice items used a 5-point Likert scale (Strongly Disagree, Disagree, Neutral, Agree, Strongly Agree), enabling quantitative measurement of adolescents' attitudes, knowledge levels, and online behaviors.

2.4.1 Physical Data Collection

Printed questionnaires were distributed directly to students during scheduled visits to each of the six colleges. Researchers were present throughout the data collection process to explain the purpose of the study, guide participants through the questionnaire, and clarify any questions. This direct engagement helped improve response quality and minimized misinterpretation of questionnaire items.

2.4.2 Data Entry

After physical collection, all completed paper responses were manually entered into Google Forms by the research team. This step ensured that all data was digitally stored in a structured and consistent format, making it suitable for systematic cleaning, organization, and analysis.

The questionnaire was reviewed for clarity, simplicity, and appropriateness before distribution to ensure accurate and honest responses from all participants.

2.5 Data Management and Analysis Strategies

After physical data collection and manual entry into Google Forms, all responses were exported into a master spreadsheet for cleaning, organization, and analysis. The following steps were undertaken to ensure data quality, accuracy, and confidentiality.

2.5.1 Data Cleaning

A total of 395 responses were initially collected through physical visits across six colleges. During the data cleaning process, 23 responses were removed for the following reasons:

- Incomplete responses with more than 20% of fields left blank
- Duplicate entries identified from the same respondent
- Respondents found to be outside the 16–20 age range

The final cleaned dataset comprised 372 valid responses, which were used for all subsequent analysis.

2.5.2 Data Organization

All variables were coded systematically (e.g. Male = 1, Female = 2; Technical = 1, non-technical = 2; college names assigned numerical codes). Responses were categorized by demographic groups to enable cross-tabulation and subgroup analysis. The master dataset was organized in spreadsheet format with each row representing one respondent and each column representing one questionnaire items.

2.5.3 Secure Storage

All physical questionnaire sheets were stored securely after data entry. The digital dataset was maintained in password-protected files throughout the research process. Personally identifiable information was separated from the analysis dataset to ensure full participant confidentiality and privacy in compliance with ethical research standards.

2.5.4 Analysis Techniques

2.5.4.1 Descriptive Statistics

Frequency counts, percentages, and distributions were calculated to summarize demographic profiles, awareness levels, and digital safety practice behaviors across the

full sample of 372 respondents. This stage of analysis has been completed and preliminary findings are presented in Chapter 3.

2.5.4.2 Cross-tabulation Analysis

Relationships between demographic variables including gender, college, faculty, and academic scheme and cybersecurity awareness and safety practice scores were examined to identify patterns and differences across subgroups. This analysis has been completed for all demographic variables and key findings are incorporated into the preliminary results.

2.5.4.3 Likert Scale Analysis

Responses were analyzed by calculating the percentage of respondents selecting Agree or Strongly Agree for each item, providing a clear and consistent indicator of awareness and practice adoption levels across all 41 questionnaire items. This stage of analysis is complete and results are reflected throughout Chapter 3.

2.5.4.4 Thematic / Pattern-Based Analysis

Behavioral patterns across demographic subgroups are currently being examined to identify common experiences, recurring themes, and deeper insights regarding cybersecurity awareness and digital safety behavior among adolescents in Dhangadhi. This stage of analysis is currently in progress and the complete findings will be incorporated into the final research report.

The overall analysis focused on interpreting results clearly and objectively, reflecting the real experiences of adolescents and highlighting key patterns in cybersecurity awareness and digital safety behavior within the study area.

Chapter 3

RESULTS AND FINDINGS

This study assessed cybersecurity awareness and digital safety practices among adolescents aged 16–20 in Dhangadhi, Nepal. A total of 395 questionnaires were physically distributed across six colleges, of which 372 valid responses were retained after data cleaning. This chapter presents the major findings organized around the study's specific objectives, beginning with the demographic profile of respondents followed by cybersecurity awareness levels, digital safety practices, influencing factors, and recommendations.

3.1 Demographic Profile of Respondents

This section presents the demographic characteristics of the 372 valid respondents, including gender, college, age, internet usage, academic background, and faculty distribution.

3.1.1 Gender Distribution

Out of 372 respondents, 193 (51.9%) identified as Male and 178 (47.8%) as Female, with 1 (0.3%) identifying as Other. The distribution reflects a near-equal gender balance, ensuring that findings are representative of both male and female adolescents in Dhangadhi.

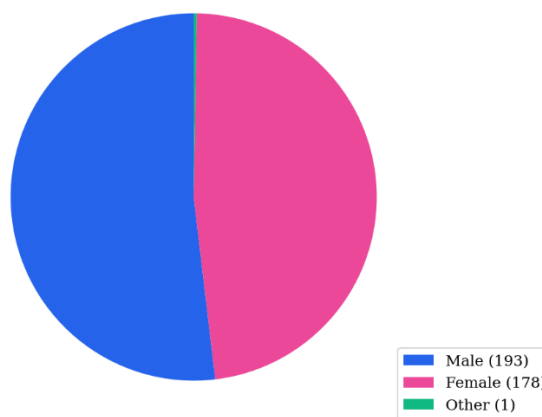


Figure 3.1 Gender Distribution of Respondents (N=372)

3.1.2 College-wise Distribution

Respondents were collected from six colleges across Dhangadhi. NAST had the highest participation with 76 respondents (20.4%), followed by AVN College (71, 19.1%) and

Alliance College (70, 18.8%). KMC College had the lowest participation with 41 respondents (11.0%), as shown in Figure 3.2.

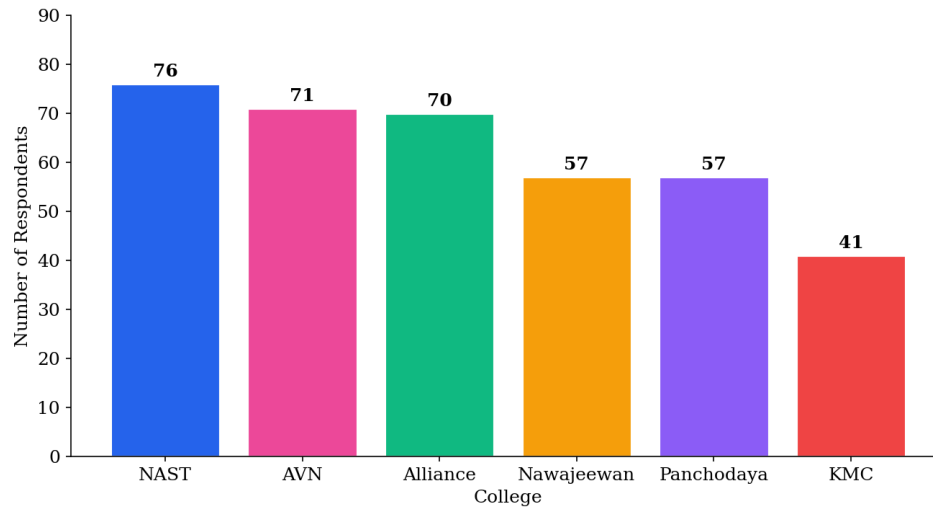


Figure 3.2 College-wise Distribution of Respondents (N=372)

3.1.3 Age Distribution

Most respondents were aged 17 years (159, 42.7%), followed by 18-year-olds (100, 26.9%). Respondents aged 16 comprised 68 (18.3%) of the sample. Only a small proportion were aged 19 (27, 7.3%) and 20 (18, 4.8%), reflecting the typical age range of +2 level students in Dhangadhi.

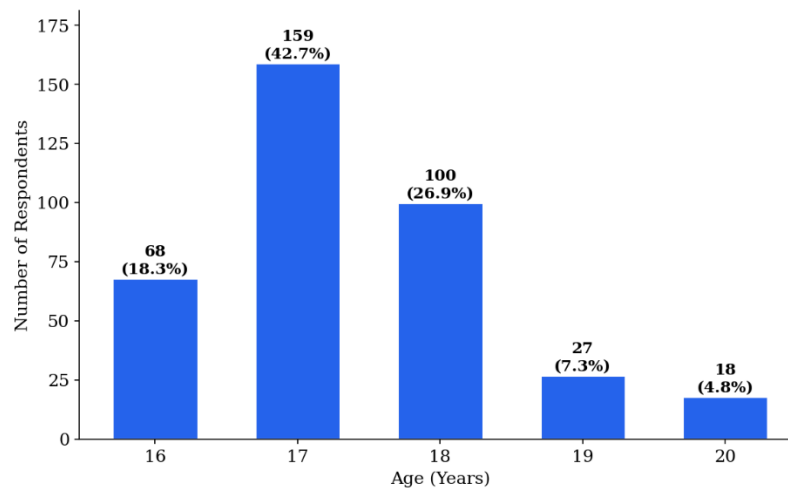


Figure 3.3 Age Distribution of Respondents (N=372)

3.1.4 Gender Distribution by College

Cross-tabulation of gender and college reveals notable variations across institutions. Nawajeewan College had considerably higher female participation (43 Female vs. 14

Male), while Alliance College showed higher male participation (52 Male vs. 18 Female). Other colleges showed relatively balanced gender distributions.

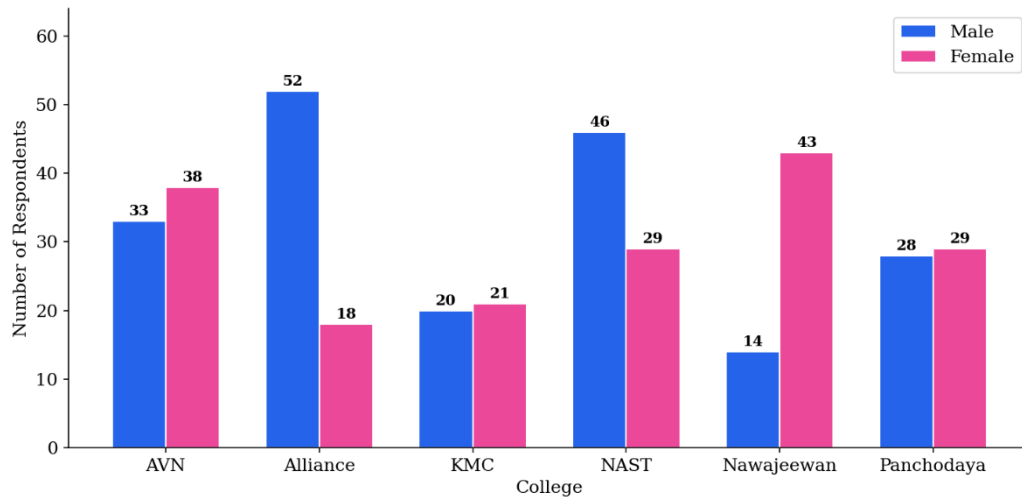


Figure 3.4 Gender Distribution by College (N=372)

3.2 Daily Internet Usage

The most common daily internet usage category was 2–4 Hours (162 respondents, 43.5%), followed by 4–6 Hours (90, 24.2%). A notable proportion (47, 12.6%) used the internet for more than 6 hours daily, indicating heavy internet dependence among a segment of adolescents. Only 73 respondents (19.6%) used the internet for less than 2 hours per day.

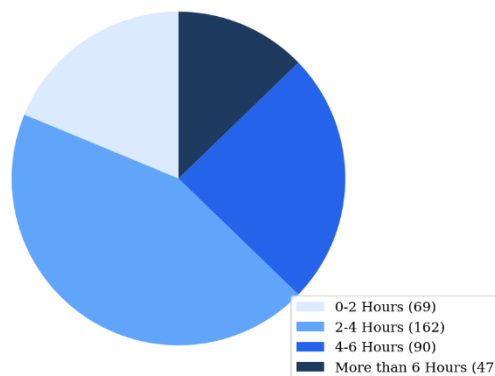


Figure 3.5 Daily Internet Usage of Respondents (N=372)

3.3 Technical vs Non-Technical Background

Out of 369 respondents with recorded scheme data, 259 (70.2%) were from non-technical backgrounds including Management, Health Science, and Bio Science, while 110 (29.8%) were from technical backgrounds including Computer Science. This

distribution is relevant as technical background may influence cybersecurity awareness and practice levels.

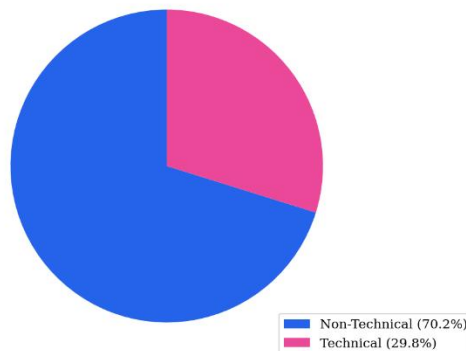


Figure 3.6 Technical vs Non-Technical Background (N=369)

3.4 Faculty-wise Distribution

Respondents came from a variety of faculties, reflecting the diverse academic backgrounds of adolescents across Dhangadhi colleges. The faculty-wise distribution provides additional context for understanding differences in cybersecurity awareness across academic streams.

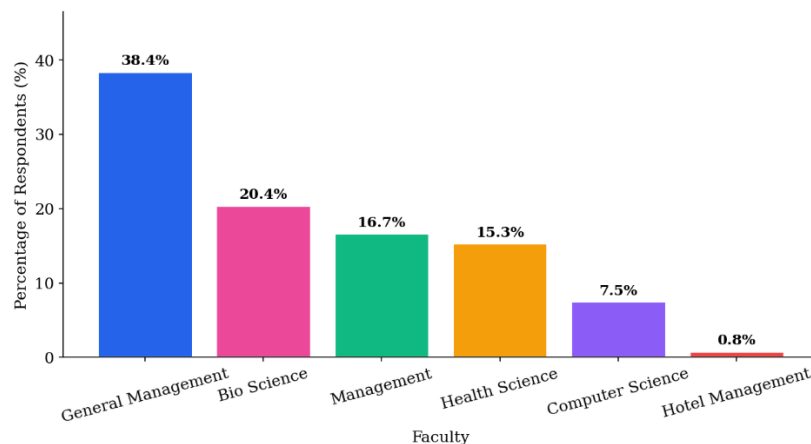


Figure 3.7 Faculty-wise Distribution of Respondents (N=372)

3.5 Cybersecurity Awareness Among Adolescents

The findings reveal a moderate-to-high level of general cybersecurity awareness among respondents, though significant gaps remain in specific areas. A large majority 91.6% recognized the importance of strong passwords, and 75.7% were aware of public Wi-Fi risks. Similarly, 73.8% understood the risks associated with clicking unknown links,

and 70.1% reported general awareness of cyber threats such as phishing, scams, and malware.

However, awareness dropped considerably in more advanced areas. Only 44.3% felt confident identifying suspicious messages, and just 53.5% reported knowledge and use of Two-Factor Authentication (2FA). These results indicate that while foundational awareness exists among adolescents in Dhangadhi, deeper technical understanding of cybersecurity threats remains limited.

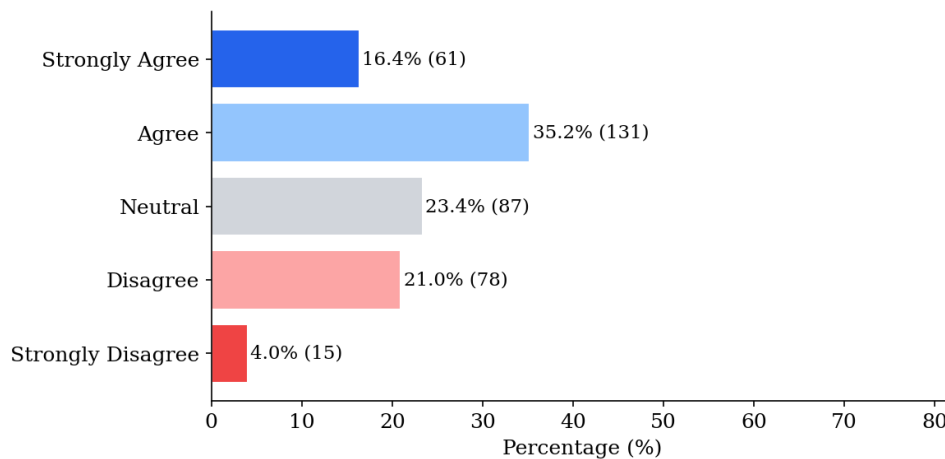


Figure 3.8 Cybersecurity Awareness Indicators Likert-Scale Distribution (N=372)

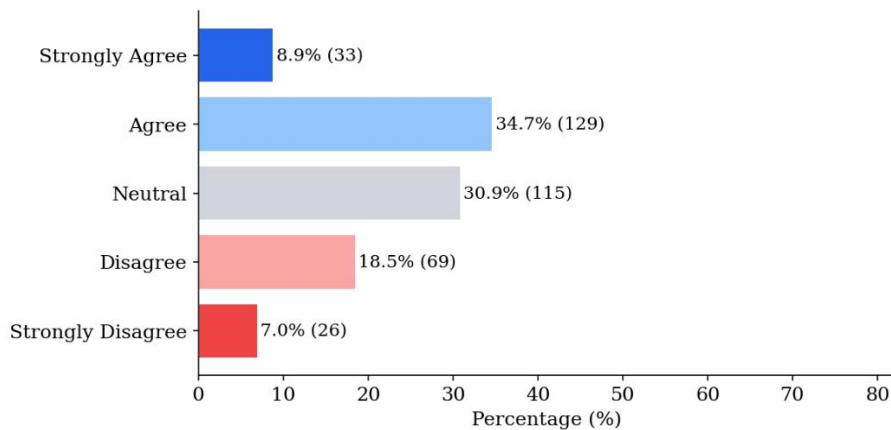


Figure 3.9 Awareness of Suspicious Messages (N=372)

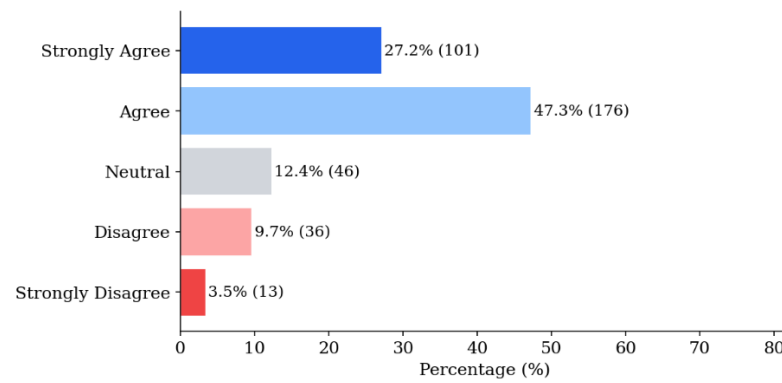


Figure 3.10 Awareness of Public Wi-Fi Risk (N=372)

3.6 Analysis of Digital Safety Practices

Despite moderate awareness levels, a noticeable gap was observed between knowledge and actual practice. On the positive side, 86.3% of respondents reported using strong passwords, 78.7% stated they do not share OTPs or sensitive information, and 74.4% reported avoiding unknown links. Additionally, 73.1% reported logging out from shared devices.

However, several unsafe practices were also documented:

Only 37.6% used antivirus software or a VPN, only 50.7% regularly updated their software and applications, 47.8% admitted to ignoring cybersecurity risks even when they were aware of them. A significant proportion accepted terms and conditions without reading them and reused passwords across multiple platforms. These findings confirm that awareness alone does not guarantee safe digital behavior, highlighting a critical knowledge-practice gap among adolescents in Dhangadhi.

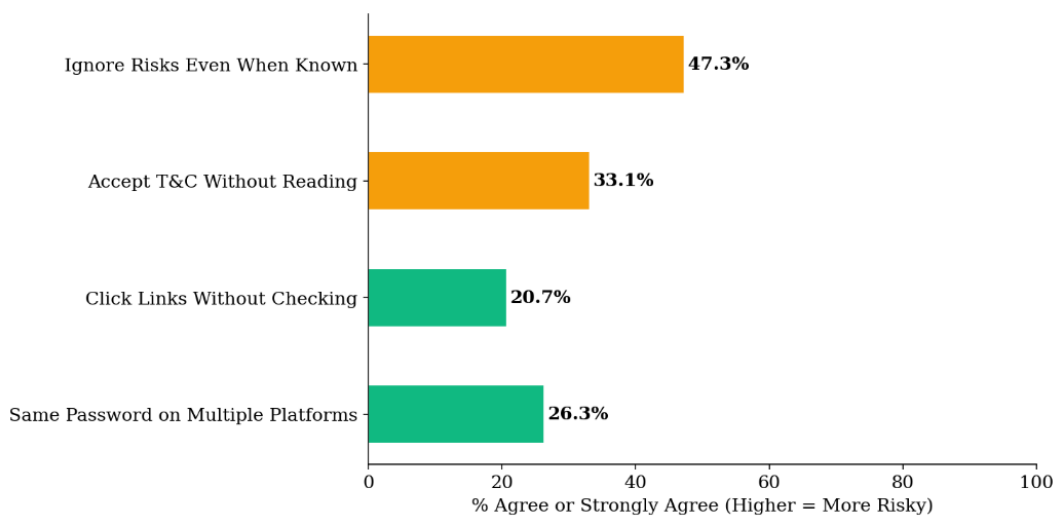


Figure 3.11 Risky Digital Behaviors Among Respondents (N=372)

3.7 Identification of Factors Influencing Awareness and Safety

The study examined how variables such as gender, college, faculty, internet usage, and technical background influence cybersecurity awareness and online safety behaviors. Respondents from technical backgrounds (Computer Science) showed comparatively higher familiarity with tools like 2FA and VPNs compared to those from non-technical streams. Adolescents who spent more time online 12.6% used the internet more than 6 hours daily reported higher exposure to cyber threats.

Notably, 49.5% of respondents reported that cyber incidents had affected either themselves or someone they know, indicating that cybersecurity risks are a real and prevalent issue in the local context and not merely a theoretical concern. Furthermore, 85.1% of all respondents affirmed that cybersecurity awareness is important, reflecting a strong community readiness for structured awareness programs.

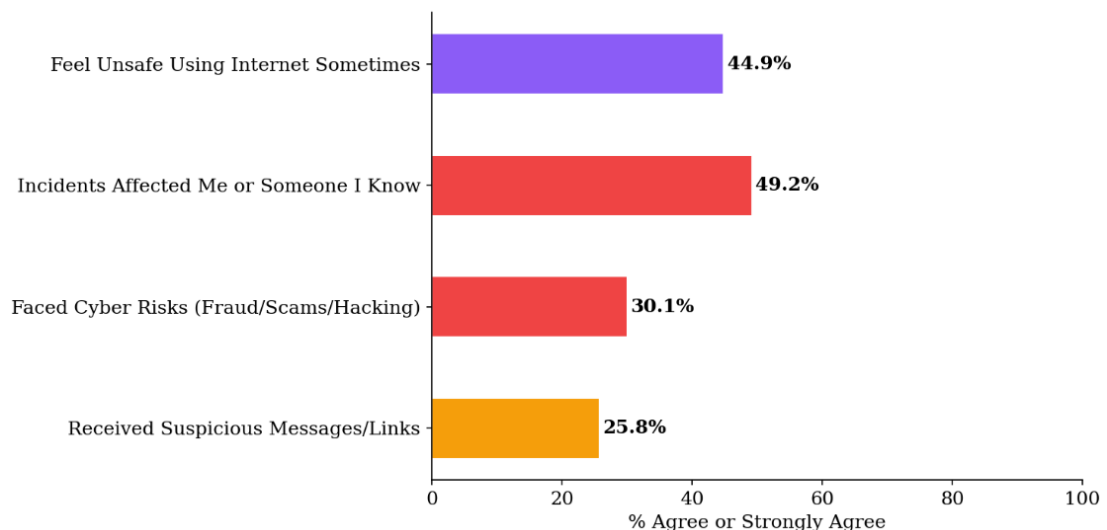


Figure 3.12 Exposure to Cyber Threats (N=372)

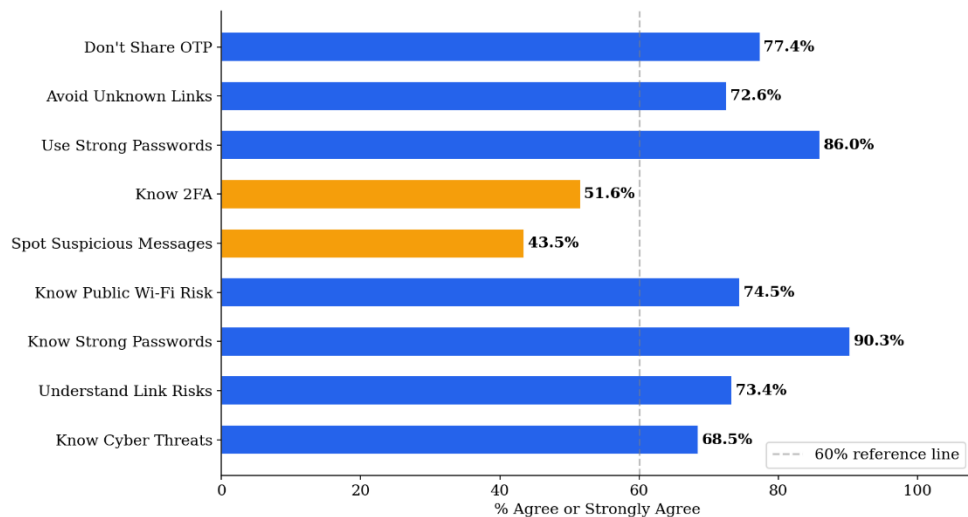


Figure 3.13 Summary Cybersecurity Awareness and Safety Practices (N=372)

3.8 Evidence-Based Recommendations for Awareness Programs

Based on the findings, the following practical strategies are recommended to improve cybersecurity awareness and digital safety among adolescents in Dhangadhi:

- Integrating short digital safety modules into school and college curricula, particularly covering 2FA, phishing recognition, and safe password management.
- Organizing peer-led cybersecurity awareness sessions in colleges, leveraging the fact that 85.1% of respondents already believe cybersecurity awareness is important.
- Conducting community workshops for parents and youth to address risky behaviors such as password reuse and ignoring software updates.
- Sharing simple, locally relevant educational content via social media platforms such as Facebook and TikTok, which are widely used by adolescents in Dhangadhi.
- Promoting the use of free and legitimate security tools such as antivirus and VPN to address the critically low adoption rate of 37.6%.

3.9 Contribution to Safer Digital Environments

This research provides actionable, evidence-based insights for educators, policymakers, and community organizations in Dhangadhi and similar semi-urban regions of Nepal. The findings directly support the design of targeted cybersecurity education interventions for adolescents aged 16–20, addressing both awareness gaps and

behavioral gaps identified through the data. With 85.1% of respondents affirming that cybersecurity awareness is important, there is clear community readiness for structured awareness programs what is needed now is institutional action to translate that willingness into safer digital practices.

Chapter 4

APPLICATION OF THE RESEARCH FINDINGS

The findings of this study have practical value for educational institutions, communities, policymakers, and public awareness initiatives in Dhangadhi, Nepal. Based on responses from 372 students, the study highlights areas where cybersecurity awareness can be strengthened and translated into safer digital practices.

4.1 Application in Educational Institutions

The study revealed that while general awareness exists among students with 91.6% recognizing the importance of strong passwords and 75.7% aware of public Wi-Fi risks significant gaps remain in more practical and technical areas. Only 37.6% of respondents used antivirus software or a VPN, and only 53.5% reported using Two-Factor Authentication (2FA). These findings indicate that current educational environments are not sufficiently addressing the practical side of cybersecurity.

Educational Institutions can apply these findings in the following ways:

- Incorporating cybersecurity awareness as a dedicated topic within existing computer science, information technology, or digital literacy subjects at both school and college levels.
- Developing simple, locally relevant teaching materials covering key threat areas identified in this study, including phishing recognition, safe password practices, 2FA setup, and responsible social media use.
- Training teachers and instructors to confidently discuss cybersecurity topics in class, particularly in non-technical stream institutions where awareness was found to be comparatively lower among the 70.2% of non-technical respondents.
- Including cybersecurity awareness in student orientation programs at the start of each academic year, ensuring that all new students receive foundational digital safety knowledge before engaging heavily with campus digital resources.
- Using the data from this study as a baseline to measure improvement in student awareness over time, allowing institutions to evaluate the effectiveness of any awareness programs they introduce.
- Since 85.1% of respondents already affirmed that cybersecurity awareness is important, educational institutions have a highly receptive student base ready to

engage with structured cybersecurity education. What is needed is a consistent, curriculum-level commitment to translating that awareness into safe and responsible digital behavior.

4.2 Application in Community-Level Programs

The findings of this research have direct and immediate application within college classrooms, where the majority of the study's respondents spend a significant portion of their daily lives. Rather than relying solely on formal curriculum changes, which may take time to implement, the research findings can be applied through practical, low-cost activities carried out directly inside existing classroom settings.

The data highlights several critical behavioral gaps that can be addressed at the classroom level. For instance, 47.8% of respondents admitted to ignoring cybersecurity risks even when they were aware of them, and only 50.7% regularly updated their software and applications. These are behaviors that can be directly discussed, challenged, and improved through guided classroom interactions.

Practical classroom-level applications include:

- Dedicating a session within computer or IT classes specifically to reviewing the findings of this research, allowing students to reflect on their own online behaviors against the data collected from their peers.
- Facilitating open class discussions where students share personal experiences with cyber threats, drawing on the finding that 49.5% have already been affected by cyber incidents either directly or through someone they know.
- Assigning hands-on in-class activities such as checking and updating privacy settings on personal devices, enabling 2FA on commonly used accounts, or practicing how to identify a phishing message.
- Encouraging students to create their own awareness content such as short posters, infographics, or social media posts that can be displayed within the classroom or shared within their college community.
- Using real statistics from this study as classroom discussion prompts, for example asking students why only 37.6% use antivirus software despite 70.1% being aware of cyber threats, encouraging critical thinking about the knowledge-practice gap.

- These classroom-based applications require no additional budget, no new infrastructure, and no external facilitators. They can be implemented immediately by existing teachers using the findings of this research as the content foundation, making them highly practical and sustainable for colleges in Dhangadhi and similar semi-urban settings across Nepal.

4.3 Application for Policymakers and Digital Literacy Campaigns

The research findings are also valuable for policymakers, government bodies, and digital literacy campaign designers in Nepal. The data reveals the economic dimension of unsafe digital behavior a significant proportion of adolescents use unverified applications and ignore software updates, partly due to limited awareness of the risks involved.

The findings support the development of:

- Policies promoting free and legitimate digital security tools, particularly antivirus software and VPNs, to address the critically low adoption rate of 37.6%.
- Government-backed digital literacy programs targeting semi-urban regions such as Dhangadhi where structured cybersecurity education remains limited despite increasing internet penetration.
- Guidelines for integrating cybersecurity awareness into the national school and college curriculum, especially for non-technical stream students who comprised 70.2% of the sample and showed comparatively lower awareness of advanced security tools.
- Ward-level and municipality-level digital safety initiatives that use the findings of this research as a needs assessment baseline for Dhangadhi and Kailali district.

4.4 Application in Public Awareness Initiatives

The study has direct applications in designing public awareness materials. The risky behaviors and awareness gaps identified through the data can be simplified into accessible formats to reach a wider audience across Dhangadhi. Local stakeholders, educational institutions, and community organizations can use the findings to create:

- Posters and flyers with simple messages such as "Do not share your OTP," "Enable Two-Factor Authentication," "Avoid clicking unknown links," and "Use trusted apps only".
- Social media content for platforms such as Facebook and TikTok, which are widely used by adolescents in Dhangadhi, promoting safe digital practices in Nepali and local languages.
- Short video clips or infographics highlighting the most critical findings for example, that nearly 1 in 2 adolescents ignores cybersecurity risks even when aware of them.
- Academic network campaigns within colleges to spread awareness using peer influence, which is particularly effective among the 16–20 age group.

These awareness materials can be distributed through community pages, public events, college notice boards, and ward-level digital literacy programs, ensuring that the research findings translate into visible, practical change in the local community.

Chapter 5

PROGRESS AGAINST APPROVED MILESTONE

The research activities have generally progressed according to the timeline approved in the proposal. Major tasks, including questionnaire development, data collection, data processing, analysis, visualization, and preparation of the midterm report, have been completed as scheduled. The progress made against each approved milestone is presented below.

Table 5.1
Progress Milestone

Activity	Scheduled Period	Status
Proposal approval	February, Weeks 3-4	Completed
Questionnaire development and pilot testing	April, Week 2	Completed
Data collection	April, Weeks 3-4	Completed
Data entry and processing	May, Weeks 1-2	Completed
Initial data analysis (data cleaning)	May, 3	completed
Data Visualization	May, Week 4	completed
Report writing(midterm)	June, Week 1	completed
Midterm defense	June, Week 2	Pending
Thematic and Pattern based analysis	June, Week 3-4	Pending
Hypothesis Analysis	July, Week 1-2	Pending
Final report writing	July Week 3-4	Pending

The research has progressed according to the approved timeline. All planned activities up to the preparation of the midterm report have been successfully completed. The only remaining milestone at this stage is the midterm defense, which is scheduled for June, Week 2.

Chapter 6

BUDGET UTILIZATION

The budget utilization for the research indicates that the project has remained within the approved financial allocation. As of the midterm stage, a total of NPR 38,000 has been utilized out of the approved budget of NPR 56,500, leaving a remaining balance of NPR 18,500 for the completion of the remaining activities. Most of the expenditure was incurred during field visits, data collection, procurement of research materials, printing, and the use of necessary software and accessories. The remaining funds are expected to support activities related to awareness and seminar dissemination, contingency requirements, and the final stages of the research.

Table 6.1
Budget Utilization

S.No.	Budget Item	Approved (NPR)	Spent to Date (NPR)	Remaining (NPR)
1.	Field work and community visits	15,000	12,000	3,000
2.	Research Materials	8,000	6,500	1,500
3.	Equipment and Accessories	4,000	3,500	500
4.	Awareness and seminar dissemination	9,500	3,000	6,500
5.	Printing & Stationery	10,000	7,000	3000
6.	Software and Tools	5,000	3,000	2000
7.	Contingency	5,000	2,000	3000
Total		56,500	37,000	19,500

Chapter 7

CHALLENGES AND MITIGATION MEASURES

Conducting research in a real-world setting inevitably involves a range of practical, logistical, and methodological challenges. This study on cybersecurity awareness and digital safety practices among adolescents aged 16–20 in Dhangadhi, Nepal encountered several such challenges across its planning, data collection, and analysis phases. Addressing these challenges effectively was essential to maintaining the academic integrity, validity, and reliability of the research outcomes. This chapter systematically documents each major challenge encountered during the study, along with the specific measures adopted by the research team to mitigate their impact and uphold the overall quality of the research process.

7.1 Obtaining Permission from Colleges

7.1.1 Challenge

One of the earliest challenges encountered was obtaining formal permission from college administrations to conduct the research within their institutions. Some colleges required prior written requests, approval from department heads, or coordination with class teachers before allowing data collection to proceed. This caused delays in the initial data collection schedule and required additional follow-up visits before questionnaire distribution could begin.

7.1.2 Mitigating Measures

The research team prepared formal permission letters on behalf of NAST, clearly stating the academic purpose of the study, the confidentiality of participant responses, and the non-invasive nature of the questionnaire. Personal visits were made to college administrations prior to the scheduled data collection dates to explain the study objectives and address any institutional concerns. This proactive approach helped establish trust with college authorities and ensured smooth, uninterrupted access to students across all six colleges Aishwarya Vidya Niketan (AVN), Alliance International Model School, Kailali Model School, Panchodaya Higher Secondary School, Navajeevan College of Nursing and Health Science, and NAST Secondary School.

7.2 Ensuring Honest and Accurate Responses

7.2.1 Challenge

Since the questionnaire addressed personal digital behaviors including potentially sensitive practices such as password reuse, ignoring security warnings, clicking unknown links, and sharing personal information online there was a considerable risk of social desirability bias. Adolescent respondents may have felt inclined to present themselves as more digitally responsible than they are, which could have compromised the accuracy and authenticity of the findings, particularly in the digital safety practices and risky behavior sections of the questionnaire.

7.2.2 Mitigating Measures

Prior to distributing questionnaires at each college, the research team verbally communicated to all participants that the questionnaire was completely anonymous, that no personally identifiable information would be recorded or disclosed, and that there were no right or wrong answers. Respondents were encouraged to answer as honestly as possible, with the assurance that their individual responses would never be shared with their college, teachers, or any other party. This transparent and reassuring approach helped create a comfortable response environment, reducing the likelihood of biased or inaccurate answers.

7.3 Managing a Large Physical Dataset

7.3.1 Challenge

Since all 395 questionnaires were collected physically across six colleges in Dhangadhi, managing, organizing, and digitizing a large volume of paper-based responses presented a significant logistical challenge. Manual data entry into Google Forms was inherently time-consuming and carried an elevated risk of data entry errors, missing values, and inconsistencies between the original paper responses and the digital dataset. Any such errors, if left unchecked, could have undermined the accuracy and reliability of the entire dataset.

7.3.2 Mitigating Measures

The research team divided the data entry workload systematically, with each team member assigned responsibility for entering responses from specific colleges to maintain accountability and consistency. A structured double-checking process was

applied, whereby completed digital entries were reviewed against the original paper questionnaires to identify and correct any discrepancies. Following data entry, a comprehensive data cleaning process was conducted, which involved removing incomplete responses, eliminating duplicate entries, and standardizing all categorical variables. This process resulted in the exclusion of 23 invalid responses and the finalization of a clean, reliable dataset of 372 valid responses used for all subsequent analysis.

7.4 Reaching Respondents Across Diverse Faculties

7.4.1 Challenge

The six colleges visited during data collection offered a wide range of academic faculties, including Computer Science, Management, Health Science, and Bio Science. Achieving balanced representation across all faculties was challenging, as certain classes were unavailable during scheduled visit times due to examinations, extracurricular activities, or teacher absences. In some instances, target classes had lower-than-expected attendance on the day of data collection, affecting the planned distribution of responses across academic streams.

7.4.2 Mitigating Measures

The research team coordinated with class teachers and college coordinators in advance to identify the most appropriate time slots for questionnaire distribution, minimizing disruption to regular academic schedules. Where specific classes were unavailable on a given visit, alternative classes from the same college and similar academic levels were approached to maintain the target response count per institution. This flexible and adaptive approach ensured broad representation across diverse academic backgrounds, ultimately resulting in a dataset comprising 70.2% non-technical stream respondents and 29.8% technical stream respondents a distribution that accurately reflects the broader student population in Dhangadhi colleges.

7.5 Limited Familiarity with Cybersecurity Terminology

7.5.1 Challenge

During physical data collection, a notable proportion of respondents particularly those from non-technical faculties such as Management, Health Science, and Bio Science expressed unfamiliarity with specific cybersecurity terminology used in the

questionnaire. Terms such as Two-Factor Authentication (2FA), Virtual Private Network (VPN), phishing, and malware were not consistently understood across all respondent groups. This created a risk of misinterpretation of questionnaire items, which could have introduced inconsistencies in the awareness and practice sections of the dataset and affected the validity of responses from non-technical respondents.

7.5.2 Mitigating Measures

The research team remained physically present throughout the data collection process at every college visit, making themselves actively available to clarify any terms or questionnaire items that respondents found confusing or unfamiliar. Brief, neutral verbal explanations of key terms were provided upon request, carefully avoiding any language that could influence or lead the respondent's answer. This on-site support significantly reduced misinterpretation of questionnaire items. For the final research report, a short glossary of key cybersecurity terms will be incorporated as an appendix to further support reader comprehension and to assist any future researchers who may replicate or build upon this study.

7.6 Ensuring Data Confidentiality and Privacy

7.6.1 Challenge

Collecting data on personal digital behaviors and online experiences from adolescent respondents raised important ethical responsibilities around data privacy and confidentiality. Ensuring that all participant data was handled, stored, and processed securely and that no individual respondent could be identified from the dataset or the published findings was both a fundamental ethical obligation and a practical challenge requiring careful management throughout the research process.

7.6.2 Mitigating Measures

All completed paper questionnaires were stored securely following data entry, with access strictly limited to the four members of the research team. The digital dataset was maintained exclusively in password-protected files, and all personally identifiable fields were removed from the analysis dataset before any processing or reporting took place. Prior to completing the questionnaire, all participants were clearly informed of these privacy protections, and informed verbal consent was obtained from every respondent before participation. No names, contact numbers, email addresses, or any

other identifying information were recorded at any stage of the data collection or entry process. These measures ensured that the research was conducted in full compliance with ethical standards for academic research involving human participants.

Chapter 8

CONCLUSION

This study set out to assess cybersecurity awareness and digital safety practices among adolescents aged 16–20 in Dhangadhi, Nepal, drawing on 372 valid responses collected through physical visits to six colleges. The findings paint a clear and consistent picture: adolescents in Dhangadhi possess a reasonably strong foundation of basic cybersecurity awareness, but this awareness does not reliably translate into safe digital behavior.

On the positive side, foundational knowledge is widespread. Most respondents (91.6%) recognized the importance of strong passwords, and most understood common risks such as unsafe public Wi-Fi (75.7%) and suspicious links (73.8%). Encouragingly, 85.1% of respondents affirmed that cybersecurity awareness matters indicating a population that is receptive, not resistant, to further education.

However, the study also surfaced a significant knowledge-practice gap. Despite high awareness of password importance, only 53.5% used Two-Factor Authentication and just 37.6% used antivirus software or a VPN. Most strikingly, 47.8% admitted to ignoring known cybersecurity risks even when aware of them directly confirming that awareness alone is insufficient to drive safe behavior. This gap was compounded by the fact that nearly half of all respondents (49.5%) reported being personally affected, or knowing someone affected, by a cyber incident, demonstrating that these risks are not abstract but actively present in adolescents' daily lives.

Differences across subgroups further enriched the findings. Students from technical streams (29.8% of the sample) showed greater familiarity with tools like 2FA and VPNs compared to their non-technical peers (70.2%), and heavier internet users (12.6% online more than six hours daily) reported greater exposure to threats suggesting that both academic background and usage intensity shape vulnerability.

Taken together, these findings lead to one central conclusion: the barrier to digital safety among adolescents in Dhangadhi is not a lack of awareness, but a failure to convert that awareness into consistent practice. This reframes the problem for educators and policymakers the priority is not simply "raising awareness" through one-off campaigns, but designing sustained, practical interventions (classroom activities, peer-led sessions, curriculum integration) that close the gap between what students know and what they do online.

Given the high baseline of willingness (85.1%) and the concrete, locally-grounded evidence this study provides, Dhangadhi's colleges are well positioned to implement targeted, low-cost cybersecurity education. Doing so would directly address the specific behavioral gaps low 2FA and antivirus adoption, risk-ignoring tendencies, and unsafe sharing practices identified through this research, contributing meaningfully to a safer digital environment for adolescents in the region.

Limitations and Future Research: As thematic/pattern analysis is still in progress, deeper qualitative insights into why adolescents disengage from safe practices despite awareness remain to be fully explored, and should be incorporated into the final report. Future studies could also expand to other districts of Kailali to test whether these findings generalize beyond Dhangadhi, and could include a follow-up survey after an awareness intervention to measure actual behavioral change.

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APPENDIX

Likert-Scale Questionnaire

SECTION A: Basic information

1. Age

☐ 16 ☐ 17 ☐ 18 ☐ 19 ☐ 20

2. Gender

☐ Male ☐ Female ☐ Other

3. Education

☐ 9th ☐ 10th ☐ 11th ☐ 12th

4. Daily internet usage

☐ 0-2 Hours ☐ 2-4 Hours ☐ 4-6 Hours ☐ More than 6 Hours

SECTION B: Digital usage and exposure

5. I spend a lot of time online daily.

☐ Strongly Disagree ☐ Disagree ☐ Neutral ☐ Agree ☐ Strongly Agree

6. I frequently use social media.

☐ Strongly Disagree ☐ Disagree ☐ Neutral ☐ Agree ☐ Strongly Agree

7. I use the internet for study, communication, and entertainment.

☐ Strongly Disagree ☐ Disagree ☐ Neutral ☐ Agree ☐ Strongly Agree

8. I often use public Wi-Fi.

☐ Strongly Disagree ☐ Disagree ☐ Neutral ☐ Agree ☐ Strongly Agree

9. I download software, audio/video/images or files regularly.

☐ Strongly Disagree ☐ Disagree ☐ Neutral ☐ Agree ☐ Strongly Agree

SECTION C: Cybersecurity awareness

10. I know the cyber threats.

☐ Strongly Disagree ☐ Disagree ☐ Neutral ☐ Agree ☐ Strongly Agree

11. I understand risks of unknown links.

☐ Strongly Disagree ☐ Disagree ☐ Neutral ☐ Agree ☐ Strongly Agree

12. I know how sensitive information can be stolen.

☐ Strongly Disagree ☐ Disagree ☐ Neutral ☐ Agree ☐ Strongly Agree

13. I know the social media privacy settings.

☐ Strongly Disagree ☐ Disagree ☐ Neutral ☐ Agree ☐ Strongly Agree

14. I know strong passwords are important.

☐ Strongly Disagree ☐ Disagree ☐ Neutral ☐ Agree ☐ Strongly Agree

15. I know public Wi-Fi is risky.

☐ Strongly Disagree ☐ Disagree ☐ Neutral ☐ Agree ☐ Strongly Agree

16. I know about computer viruses.

☐ Strongly Disagree ☐ Disagree ☐ Neutral ☐ Agree ☐ Strongly Agree

17. I can spot suspicious messages.

☐ Strongly Disagree ☐ Disagree ☐ Neutral ☐ Agree ☐ Strongly Agree

18. I know about ecommerce and online transaction.

☐ Strongly Disagree ☐ Disagree ☐ Neutral ☐ Agree ☐ Strongly Agree

SECTION D: Digital literacy

19. I know how to change privacy setting.

☐ Strongly Disagree ☐ Disagree ☐ Neutral ☐ Agree ☐ Strongly Agree

20. I can identify fake software/Apps, websites and weblinks.

☐ Strongly Disagree ☐ Disagree ☐ Neutral ☐ Agree ☐ Strongly Agree

21. I know how to secure my accounts.

☐ Strongly Disagree ☐ Disagree ☐ Neutral ☐ Agree ☐ Strongly Agree

22. I know how to protect my personal information.

☐ Strongly Disagree ☐ Disagree ☐ Neutral ☐ Agree ☐ Strongly Agree

SECTION E: Digital safety practices

23. I use strong passwords.

☐ Strongly Disagree ☐ Disagree ☐ Neutral ☐ Agree ☐ Strongly Agree

24. I avoid unknown links.

☐ Strongly Disagree ☐ Disagree ☐ Neutral ☐ Agree ☐ Strongly Agree

25. I don't share OTP, passwords and sensitive information to other.

☐ Strongly Disagree ☐ Disagree ☐ Neutral ☐ Agree ☐ Strongly Agree

26. I update software/apps regularly.

☐ Strongly Disagree ☐ Disagree ☐ Neutral ☐ Agree ☐ Strongly Agree

27. I use two-factor authentication (2FA) and secure privacy settings.

☐ Strongly Disagree ☐ Disagree ☐ Neutral ☐ Agree ☐ Strongly Agree

28. I log-out the account from shared devices.

☐ Strongly Disagree ☐ Disagree ☐ Neutral ☐ Agree ☐ Strongly Agree

29. I avoid unsafe downloads.

☐ Strongly Disagree ☐ Disagree ☐ Neutral ☐ Agree ☐ Strongly Agree

30. I use antivirus and secure VPN.

☐ Strongly Disagree ☐ Disagree ☐ Neutral ☐ Agree ☐ Strongly Agree

31. I share cyber knowledge to people who don't know about it.

☐ Strongly Disagree ☐ Disagree ☐ Neutral ☐ Agree ☐ Strongly Agree

SECTION F: Behavior gap

32. Even if I know risks, I ignore them.

☐ Strongly Disagree ☐ Disagree ☐ Neutral ☐ Agree ☐ Strongly Agree

33. I keep same password on multiple platforms.

☐ Strongly Disagree ☐ Disagree ☐ Neutral ☐ Agree ☐ Strongly Agree

34. I click on links without checking them.

☐ Strongly Disagree ☐ Disagree ☐ Neutral ☐ Agree ☐ Strongly Agree

35. I accept terms and conditions of software/apps without studying properly.

☐ Strongly Disagree ☐ Disagree ☐ Neutral ☐ Agree ☐ Strongly Agree

SECTION G: Attitude

36. Cybersecurity is not important for me.

☐ Strongly Disagree ☐ Disagree ☐ Neutral ☐ Agree ☐ Strongly Agree

SECTION H: Risk exposure and experience

37. I receive suspicious messages or links.

☐ Strongly Disagree ☐ Disagree ☐ Neutral ☐ Agree ☐ Strongly Agree

38. I have faced cyber risks (fraud, scams, hacking).

☐ Strongly Disagree ☐ Disagree ☐ Neutral ☐ Agree ☐ Strongly Agree

39. Cyber incidents affected me or someone I know.

☐ Strongly Disagree ☐ Disagree ☐ Neutral ☐ Agree ☐ Strongly Agree

40. I feel unsafe using the internet sometimes.

☐ Strongly Disagree ☐ Disagree ☐ Neutral ☐ Agree ☐ Strongly Agree

41. Cybersecurity awareness is important.

☐ Strongly Disagree ☐ Disagree ☐ Neutral ☐ Agree ☐ Strongly Agree